The Hong Kong Polytechnic University

Subject Description Form

Subject Code	AP30018
Subject Title	Acoustics Principles and Intelligent Applications
Credit Value	3 credits
Level	3
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: AP20017 Mechanics and Robotic Motion Exclusion: AP30001 Applied Acoustics
Objectives	To provide students with an introductory overview to a wide range of acoustics terminology and phenomena, and the use of artificial intelligence and machine learning techniques in related topics, including: the theory and principles of acoustics; environmental noise and vibration measurements; principles of ultrasound; transducers and applications in medicine and industry.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) apply the basic theory and principles of audible sound and ultrasound in calculations and measurements; (b) describe different causes of noise problems and find remedies to control the noise; (c) calculate the reverberation time of a room and suggest how to design a room with optimal reverberation time; (d) describe the structure of ultrasonic transducer and their applications in intelligent condition monitoring systems; (e) describe various types of devices for sound recording and reproduction, and the process for sound recognition.
Subject Synopsis/ Indicative Syllabus	**Principles of acoustics**: behaviour of sound waves in air; radiation of sound from a point source in the free field; acoustic impedance; room acoustics; vibration. **Noise measurements and control**: the nature of noise; noise criteria and rating curves; other standards of noise control (Leq etc.); noise control programmes. **Ultrasound transduction and intelligent applications**: piezoelectric materials; transducer construction; field patterns at near field and far field; medical and intelligent applications such as non-destructive testing, acoustics analytics. **Sound recording and recognition**: loudspeakers; microphones; storage media; audio file formats; sound recognition.
Teaching/Learning Methodology	**Lecture**: The concepts related to selected topics in acoustics will be explained. Examples will be used to illustrate the concepts and ideas in the lectures. The students are free to request help. Assignment sets will be given to assess the learning progress of students.

<table>
<tr><td></td><td>Tutorial: Various teaching and learning activities will be conducted in tutorial sessions to consolidate the teaching in lectures. Students will work on problem sets in the tutorials, which provide them opportunities to apply the knowledge gained in lectures. Case studies and presentation on specific acoustic applications allows students to develop a deeper understanding of the subject in relation to engineering science and industrial practices.

Laboratory: Students will work in groups and conduct experiments, both physical and computational, under the guidance of the teaching staff. They are required to analyze their experimental results and complete laboratory reports during the laboratory sessions.</td></tr>
<tr><td>Assessment Methods in Alignment with Intended Learning Outcomes</td><td>
<table>
<tr><td rowspan="2">Specific assessment methods/tasks</td><td rowspan="2">% weighting</td><td colspan="5">Intended subject learning outcomes to be assessed (Please tick as appropriate)</td></tr>
<tr><td>a</td><td>b</td><td>c</td><td>d</td><td>e</td></tr>
<tr><td>1. Continuous assessment</td><td>40</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr>
<tr><td>2. Examination</td><td>60</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr>
<tr><td>Total</td><td>100</td><td colspan="5"></td></tr>
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Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:

Continuous assessment includes assignments, laboratory report and tests which aim at checking the progress of students throughout the course, assisting them in self monitoring their performances. Laboratory sessions are designed to provide hands-on experiences to the students on specialized acoustical instruments and techniques. The final examination will be used to assess the knowledge acquired by the students, as well as to determine the extent in which they have achieved the intended learning outcomes.</td></tr>
<tr><td>Student Study Effort Expected</td><td>
<table>
<tr><td>Class contact:</td><td></td></tr>
<tr><td>▪ Lecture & Tutorial</td><td>36 Hrs.</td></tr>
<tr><td>▪ Laboratory</td><td>9 Hrs.</td></tr>
<tr><td>Other student study effort:</td><td></td></tr>
<tr><td>▪ Self-study</td><td>75 Hrs.</td></tr>
<tr><td>Total student study effort</td><td>120 Hrs.</td></tr>
</table></td></tr>
<tr><td>Reading List and References</td><td>Miles, “Physical Approach to Engineering Acoustics” 2020, Springer.

Jáuregui-Correa and Lozano Guzman, “Mechanical Vibrations and Condition Monitoring”, 2020, Academic Press.</td></tr>
</table>

	Camastra and Vinciarelli, “Machine Learning for Audio, Image and Video Analysis: Theory and Applications”, 2nd Edition, 2015, Springer.